How Much of a Time Series Survives Its Image? A Cross-Domain Comparative Analysis of the Reconstruction Capacity of Image-Based Representations

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Abstract

Image-based representations have been widely studied for time-series classification, while recent work has also demonstrated their use for image-based synthesis and imputation. Yet successful generation does not establish what different encodings preserve or whether their images can be reliably decoded into signals. We investigate this question for binary, unsigned and signed recurrence plots (RP), Gramian angular summation and difference fields (GASF/GADF), and Markov transition fields (MTF), using the invertible-by-design XIRP as a reference. Alongside established inverses, we introduce an image-only geometric decoder for signed RP, a restricted-range GADF decoder, and an approximate MTF reconstruction procedure.

Across seven datasets from six domains, we evaluate temporal and spectral fidelity, robustness, computational cost and downstream classification utility, distinguishing encoding losses from 8-bit storage effects. Under stated conditions, signed RP, GASF, restricted-range GADF and XIRP allow numerically exact reconstruction from floating-point matrices; storage introduces addi-

tional error, particularly for ill-conditioned RP states. Unsigned RP retains a global-reflection ambiguity, whereas binary RP and MTF are intrinsically many-to-one.

These findings establish a reproducible comparison of reconstruction capabilities and limitations, informing future research on generating time series through image-domain models.
Keywords: Time-series imaging, Reconstruction, Recurrence plots, Gramian angular fields, Markov transition fields, Invertibility, Comparative analysis

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1. Introduction

Image-based representations transform time series into structured matrices that can be processed using techniques originally developed for images. Recurrence plots (RP), Gramian angular fields (GAF) and Markov transition fields (MTF) have been widely explored for time-series classification and related tasks [1, 2, 5, 6]. In classification, a representation need only preserve information relevant to the target task; reconstructing the original waveform is not necessarily required. However, applications that aim to recover or synthesise temporal signals introduce a different requirement: the encoded images must retain sufficient information to reconstruct meaningful time series.

Recent research has demonstrated the feasibility of exploiting image-based generative models for temporal data. Huang et al. [3] used Gramian angular representations with generative adversarial networks for traffic-data imputation. More recently, Hellermann and Lessmann [4] introduced the extended intertemporal return plot (XIRP), an invertible-by-design represen-

tation for time-series generation through image-based GANs. Their results demonstrated competitive predictive utility against specialised generative approaches and highlighted the importance of the inversion procedure. These developments establish image-based time-series generation as a promising research direction. Nevertheless, successful synthesis using particular representations does not establish the reconstruction capabilities of other encodings, nor the conditions under which their information can be recovered.

Understanding these properties requires distinguishing three aspects: the information intrinsically preserved by an encoding, the ability of a decoder to recover that information, and the stability of reconstruction after image storage or perturbation. This distinction is particularly relevant for representations whose inversion is ambiguous or has received comparatively limited attention. Moreover, accurately reconstructing an image produced by a known encoder does not guarantee that an arbitrary image generated by a learned model represents a valid time series. Characterising reconstruction is therefore a necessary preliminary investigation, rather than a demonstration of generative performance.

In this work, we establish a common framework for evaluating the reconstruction capabilities of image-based representations built from relationships between temporal samples or states. We examine binary, continuous unsigned and signed RP; Gramian angular summation and difference fields (GASF/GADF); and MTF. XIRP is included as an invertible-by-design reference developed for generative applications. Time-frequency and time-scale representations, such as spectrograms and scalograms, are outside the scope of this study.

We combine established reconstruction methods with new image-only procedures: a geometric decoder for signed RP, an anchored decoder for GADF under restricted angular normalisation, and an approximate representative-trajectory decoder for MTF. For recurrence plots, we distinguish the contribution of the encoding from that of the reconstruction algorithm through comparisons with the approaches of Hirata et al. [12] and Sipers et al. [14]. All methods are evaluated under a common 8-bit image-storage protocol, reconstructing normalised signals from the stored images alone, with shared calibration used only to recover physical units (Figure 1).

We address four research questions:

- *RQ1 — Recoverability.* What information can be recovered from each representation, in both the time and frequency domains, and how much is lost through encoding and 8-bit storage?
- *RQ2 — Reconstruction methods.* Under which conditions are the proposed decoders exact, and how do the encoding and decoding stages separately influence recurrence-plot reconstruction?
- *RQ3 — Robustness and stability.* How do image perturbations, quantisation and signal characteristics, including sampling rate and state-space conditioning, affect reconstruction fidelity?
- *RQ4 — Computational and downstream utility.* What are the computational costs of reconstruction, and to what extent do reconstructed signals preserve the performance of classifiers trained on original data?

The main contributions of this study are:

- *A reproducible reconstruction benchmark.* We systematically evaluate temporal image representations across seven public datasets from six domains, using a common protocol covering temporal and spectral fidelity, robustness, computational cost and downstream classification utility. This allows intrinsic representation losses to be distinguished from storage-related errors.

- *New reconstruction procedures and theoretical characterisation.* We introduce a geometric image-only decoder for signed RP, together with sufficient conditions for exact recovery, an anchored decoder for restricted-range GADF, and an approximate reconstruction procedure for MTF. Controlled RP comparisons further separate the effects of directional information, distance preservation and decoder design.

- *An empirical foundation for image-based time-series generation.* We identify which representations admit reliable reconstruction, the ambiguities and limitations they retain, and the conditions affecting reconstruction stability. These findings provide a methodological basis for future generative studies, without assuming that reconstruction fidelity alone determines the quality of synthetic time series.

2. Related work

Recurrence plots show when a dynamical system revisits a region of its state space [5, 6]. The conventional RP is binary; continuous variants keep the distance between states, and the signed variant of Lu and Tong also keeps the direction of the displacement [11]. Wang and Oates proposed GASF,

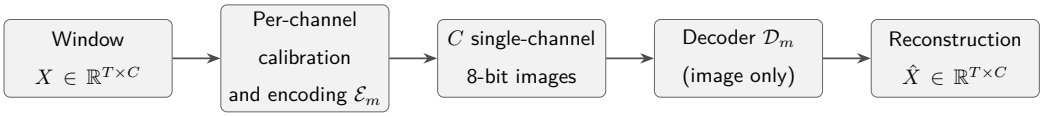


Figure 1: Controlled reconstruction. Each channel is calibrated, encoded, stored as an 8-bit image and decoded from the image alone; the shared per-channel calibration returns it to physical units. Decoding the floating-point operator instead of the image isolates the algorithmic error from the storage error.

GADF and MTF as inputs to convolutional networks [1, 2], and these representations have been applied to activity recognition [7], anomaly detection [8] and imputation [9]. XIRP was introduced for time-series generation with image-based generative adversarial networks, with the explicit aim of making generated images invertible [4].

Inversion has been studied separately for each representation. For binary RPs, Hirata et al. approximate distances between states by shortest paths in a recurrence graph and recover the states by multidimensional scaling [12, 23]. For unthresholded RPs of delay embeddings, Sipers et al. give spectral conditions under which the RP determines the signal up to its mean and sign, and a reconstruction algorithm tested under disturbances and on multi-level RPs [13, 14]. Both lines of work take an unsigned input, thresholded or not, so the sign of the series remains undetermined; neither uses the directional sign of the signed RP. GASF is read from its diagonal, after restoring damaged images with denoising autoencoders [1, 9]; approximate recovery from MTF has been reported to be difficult [1]; XIRP is inverted from its diagonal or by averaging the proposals of all its columns [4]. We are not aware of a decoder that uses the sign of the signed RP to fix the orientation from a stored image, of a GADF decoder that resolves its branch ambiguity and its lost angular

offset from the image alone, or of a comparison of these representations under a common protocol that separates representation loss from storage loss or, for recurrence plots, the contribution of the encoding from that of the decoder.

3. Mathematical formulation

3.1. Encoding, storage and decoding

Let $x = (x_1, \ldots, x_T)$ be one channel of a window. Every method m applies a calibration $\mathcal{N}$ to a method-specific normalised domain, an encoder $\mathcal{E}_m$, a fixed 8-bit storage map $\mathcal{Q}_m$ and a decoder $\mathcal{D}_m$:

$$\hat{x} = \mathcal{N}^{-1}\Big(\mathcal{D}_m(\mathcal{Q}_m(\mathcal{E}_m(\mathcal{N}(x)))); \, x_{\max}, x_{\min}\Big).$$

The calibration $(x_{\max}, x_{\min})$ is shared by all methods and is the only per-channel information used to return to physical units. We keep three properties apart: *identifiability*, whether $\mathcal{E}_m$ is injective on the normalised domain, possibly up to a stated symmetry; *exactness* of a given decoder, whether $\mathcal{D}_m \circ \mathcal{E}_m$ is the identity under stated hypotheses, so that all error is due to $\mathcal{Q}_m$; and *stability*, how much the error grows after $\mathcal{Q}_m$ or other perturbations of the image. A representation is *many-to-one* when distinct normalised series share the same matrix; no decoder can then be exact for all series.

3.2. Recurrence plots

The three RP encodings are built on the adjacent states

$$s_i = (x_i, x_{i+1}) \in \mathbb{R}^2, \qquad i = 1, \ldots, T-1,$$

a two-dimensional delay embedding with unit delay [10], and differ in what they keep of the displacements $s_i - s_j$. The *binary* RP keeps only whether

two states are close [5, 6], $B_{ij} = \mathbb{1}[\|s_i - s_j\|_2 < \varepsilon]$; it is many-to-one, and any reconstruction from it has to approximate distances from neighbourhoods. The *unsigned* RP, also called an unthresholded or global RP [6, 13], keeps the distances $D_{ij} = \|s_i - s_j\|_2$. As a Euclidean distance matrix it determines the states up to a translation and an orthogonal transformation of the plane, and the series x and its reflection $-x$ have the same D, so it is at best identifiable up to sign. It is stored as $D/\max_{ij} D_{ij}$.

The *signed* RP of Lu and Tong [11] adds whether each displacement follows a reference direction. With $v = (1, 1)/\sqrt{2}$ and $\hat{u}_{ij} = (s_i - s_j)/\|s_i - s_j\|_2$,

$$R_{ij} = \sigma_{ij}\|s_i - s_j\|_2, \qquad \sigma_{ij} = \begin{cases} +1, & \|s_i - s_j\|_2 = 0 \text{ or } \hat{u}_{ij} \cdot v \geq \cos(\pi/4), \\ -1, & \text{otherwise.} \end{cases} \tag{1}$$

Its magnitude is the unsigned RP, $|R| = D$. A pair is *sign-informative* when its displacement lies within $\pi/4$ of $\pm v$; then $\sigma_{ij} = -\sigma_{ji}$, while displacements closer to the anti-diagonal receive -1 in both orders. The convention that marks only displacements within $\pi/4$ of $-v$ differs only on non-informative pairs and carries the same information. The signed RP is stored with a per-window affine map; its zero diagonal marks the level of $R = 0$ in the image (Section 4.1).

Exact recovery needs a recoverable geometry, an overlap that leaves only x and $-x$, and a sign that tells them apart. The first two concern the unsigned geometry, studied by Sipers et al. for delay embeddings [13]; the third is what the sign adds.

Proposition 1 (Signed RP on exact input)**.** *Let $S \in \mathbb{R}^{(T-1)\times 2}$ have rows s_i, and assume*

(a) *the centred states span* $\mathbb{R}^2$*, i.e.* $\sigma_2(S - \bar{S}) > 0$*;*

(b) *for* $i = 1, \ldots, T-2$*,* x *satisfies neither* (R) $x_i + x_{i+2} = \kappa$ *nor* (F) $\alpha(x_{i+2} - x_i) = \beta x_{i+1} + \kappa$*, for any constants* κ *and* $(\alpha, \beta) \neq (0, 0)$*;*

(c) *at least one pair of states is sign-informative.*

Then the decoder of Section 4.2 applied to the floating-point signed RP returns x *on the normalised domain. Without (c),* x *and* $-x$ *have the same signed RP, so no decoder that uses the image alone can tell them apart. Under (a), condition (b) is equivalent to one of the matrices* H_η *of step 3 having a simple zero eigenvalue and the other being positive definite, so it can be checked from the image; it holds, in particular, whenever the centred triples* (x_i, x_{i+1}, x_{i+2})*,* $i = 1, \ldots, T-2$*, have rank three.*

Proof. By (a), classical MDS applied to $|R|$ returns Y with $Y = SP_0 + \mathbf{1}t^\top$ for some $P_0 \in O(2)$ [15], so every candidate configuration is $Z = SP + \mathbf{1}c^\top$ with $P \in O(2)$. Writing (a_i, b_i) for the rows of Z, the residual $b_i - a_{i+1}$ is constant in i if and only if $P_{12}x_i + (P_{22} - P_{11})x_{i+1} - P_{21}x_{i+2}$ is. For a rotation, $P = \left(\begin{smallmatrix} c & s \\ -s & c \end{smallmatrix}\right)$, this reads $s(x_i + x_{i+2}) = \text{const}$; for a reflection, $P = \left(\begin{smallmatrix} c & s \\ s & -c \end{smallmatrix}\right)$, it reads $s(x_i - x_{i+2}) - 2c\,x_{i+1} = \text{const}$. Under (b) the only solutions are $s = 0$ with P a rotation, i.e. $P = \pm I$, so the overlap is satisfied exactly by the states of x and $-x$ alone, and one of the two matrices H_η of step 3 has a simple zero eigenvalue, whose eigenvector, taken with both signs, yields these two candidates. Conversely, (R) makes every rotation a solution, so one H_η vanishes, and (F) gives a reflection solution, a zero eigenvalue of the other parity; this proves the equivalence. In step 4, x reproduces R exactly. Any candidate that also reproduces R, up to the positive scale removed by the normalisation, has states with distance matrix proportional to $|R|$ that satisfy

the overlap by construction, and is therefore $\pm x$ up to scale and translation, which the final range map removes. Reflecting the series maps $s_i - s_j$ to $s_j - s_i$, so σ_{ij} becomes σ_{ji}: on a sign-informative pair the entry changes sign, on a non-informative pair it does not. Hence the signed RPs of x and $-x$ differ if and only if (c) holds, and under (c) the selected candidate is x. □

Neither (b) nor (c) follows from (a). A sinusoid at a quarter of the sampling rate, $x_i = \sin(\pi i/2 + \phi)$, satisfies (R): every rotation then meets the overlap, and the phases $\phi = 0.2$ and $\phi = 0.25$ give the same signed RP, so identifiability can be lost altogether. Condition (b) is sufficient, not necessary: under (F) the additional reflection can still be discarded by the sign. The series $x = (1, -1.05, 1.10, -1.15, 1.20, -1.25, 1.30, -1.35)$ satisfies (a) and (b), but every displacement lies close to the anti-diagonal, so no pair is informative and x and $-x$ have the same signed RP. In the benchmark, all 61 083 channel series have a sign-informative pair, with a smallest share of 6%, and the floating-point decoder recovered every one of them to numerical precision (Section 6.1), so neither failure affected the results.

Remark 1 (Stability after 8-bit storage)**.** *Proposition 1 concerns exact input. Storage perturbs $|R|$ by at most half a quantisation step relative to its range; by the Davis–Kahan theorem, the resulting error of the recovered two-dimensional subspace is bounded by the perturbation divided by the gap between the second and third eigenvalues of the centred Gram matrix [24], which shrinks as σ_2/σ_1 of the centred states tends to zero. We therefore use σ_2/σ_1 as an indicator of conditioning. It does not cover errors in the choice of orientation, which depend on the overlap residual and on the sign comparison.*

3.3. Gramian angular fields

The Gramian angular fields [1] map each normalised sample to an angle, $\phi_i = \arccos \tilde{x}_i$, and store

$$G^{\mathrm{S}}_{ij} = \cos(\phi_i + \phi_j), \qquad G^{\mathrm{D}}_{ij} = \sin(\phi_i - \phi_j).$$

The diagonal of GASF, $\cos(2\phi_i)$, determines $\tilde{x}_i$ only up to sign on $[-1, 1]$, but uniquely on $[0, 1]$, which we use: there $\phi_i \in [0, \pi/2]$ and $\tilde{x}_i = \cos\!\left(\frac{1}{2} \arccos G^{\mathrm{S}}_{ii}\right)$, so GASF is exactly invertible from its diagonal.

GADF is antisymmetric with a zero diagonal: it stores only relative angles. Inverting it requires the principal branch of arcsin to return $\phi_i - \phi_j$ and the lost global offset to be restored. If each window is scaled to the symmetric interval $[-\alpha, \alpha]$,

$$\max_{i,j} |\phi_i - \phi_j| = \pi - 2 \arccos \alpha < \frac{\pi}{2} \iff \alpha < \frac{1}{\sqrt{2}},$$

so with $\alpha = 0.7$ no entry leaves the principal branch, and since every non-constant window reaches both bounds $\pm\alpha$ after scaling, the offset can be fixed from the image itself (Section 4.4). Exactness therefore holds for this restricted, known normalisation; it is not a property of GADF in general, and images encoded on a wider angular range keep the branch ambiguity.

3.4. Markov transition fields

MTF assigns every sample to one of Q quantile states q_t, estimates the transition matrix $W_{pq} = P(q_{t+1} = q \mid q_t = p)$ and sets $M_{ab} = W_{q_a q_b}$ [1]. Samples in the same state are indistinguishable, the image values do not fix the numerical order of the states, and different series with the same state sequence produce the same image. MTF is therefore many-to-one: no decoder

can return every series exactly, and decoding means choosing a representative trajectory.

3.5. Extended intertemporal return plot

XIRP [4] is defined for a strictly positive series as $R_{ij} = \log(x_i/x_j)$ for $i \neq j$ and $R_{ii} = x_i$. Its diagonal is the series, so it is exactly invertible by construction; every column j also proposes $x_i = x_j \exp(R_{ij})$, a redundancy that inversion by mean averages. Positivity is obtained by mapping each channel to $[1, 2]$ with the shared calibration.

Table 1 summarises what each image determines and how it is decoded, and Figure 2 shows the stored images of one window of each dataset.

4. Decoders

4.1. Storage protocol

Each matrix is written to $[0, 255]$ by a fixed or per-window affine map. GASF and GADF are read back with the fixed range $[-1, 1]$, MTF with $[0, 1]$, and XIRP with the fixed ranges of its diagonal, $[1, 2]$, and off-diagonal, $[-\log 2, \log 2]$. The signed RP keeps a per-window map; because its diagonal is zero, subtracting the median diagonal pixel recovers the signed matrix up to a positive scale, which MDS and the final range map absorb. No decoder uses per-window metadata.

4.2. Geometric decoder on the signed RP

1. Recover $R^{\text{obs}} = \text{PNG} - \text{median}_i \, \text{PNG}_{ii}$ and $D_{ij} = \frac{1}{2}(|R^{\text{obs}}_{ij}| + |R^{\text{obs}}_{ji}|)$.
2. Classical MDS [15]: with $J = I - \mathbf{1}\mathbf{1}^\top/(T-1)$, the two leading eigenpairs of $-\frac{1}{2} J D^{\circ 2} J$ give $Y = V_2 \Lambda_2^{1/2}$.

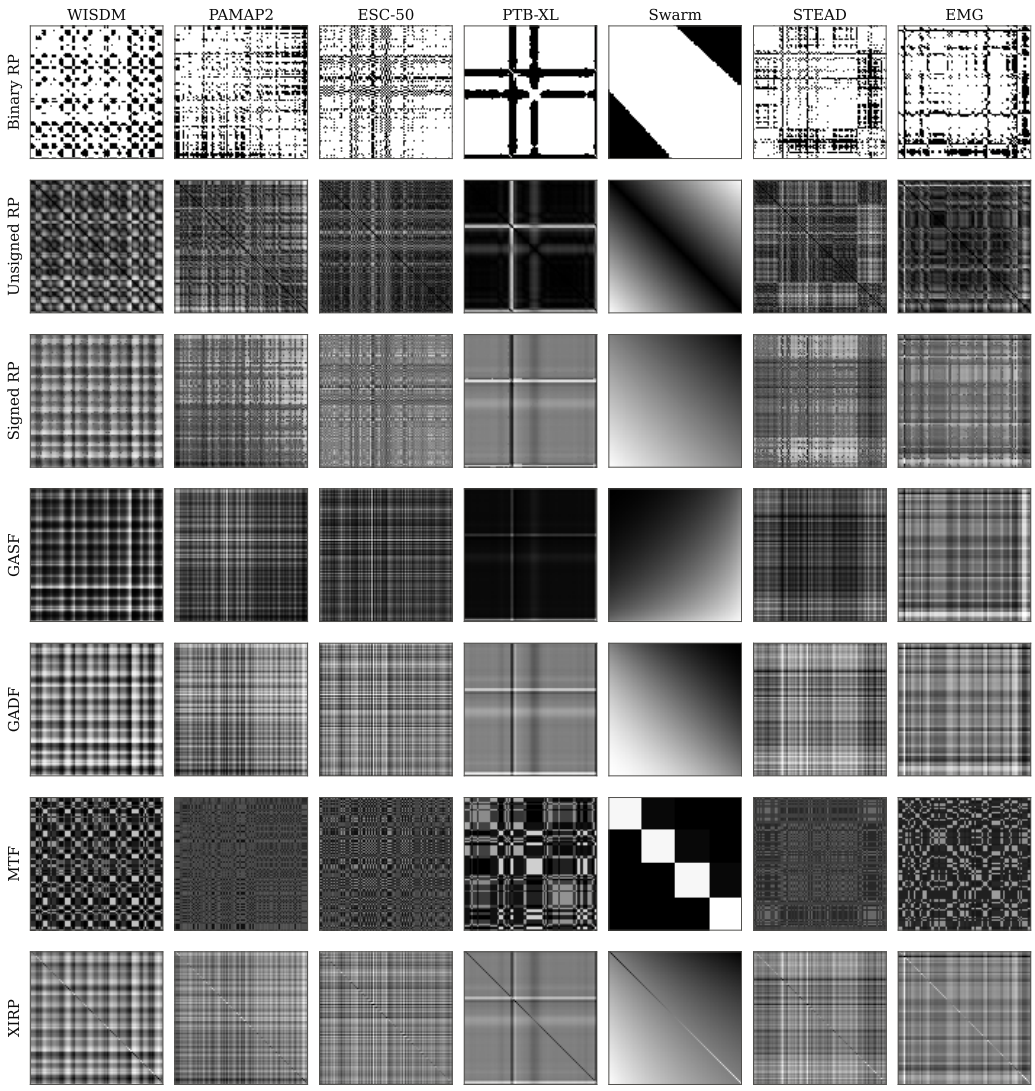


Figure 2: Stored 8-bit images ($T = 128$) of one channel of one window per dataset, in grey levels as stored (0 black, 255 white). The binary RP uses a recurrence rate of 0.75. Each column is decoded in the corresponding row of Figure 3.

Table 1: What each image determines, when exact recovery is possible, and the decoders compared. All decoders use the image alone on the normalised domain.

Encoding	Image determines	Exact recovery	Decoders (origin)
Binary RP	Recurrence neighbourhoods	No (many-to-one)	Hirata, after thresholding the unsigned RP
Unsigned RP	State geometry	Up to sign	Hirata on the thresholded image [12]; Sipers et al., circular variant [14]; geometric decoder without sign (this work)
Signed RP	State geometry and directional sign	Under Prop. 1	Hirata and Sipers with sign selection (adapted); geometric decoder (this work)
GASF	Angles on $[0, \pi/2]$	Yes, on $[0, 1]$	Diagonal inverse [1]
GADF	Angular differences	If scaled to $[-\alpha, \alpha]$, $\alpha < 1/\sqrt{2}$	Restricted branch and boundary anchor (this work)
MTF	Quantile states and transitions	No (many-to-one)	Fiedler-ordered representative (this work)
XIRP	Series on the diagonal	Yes	Inversion by mean [4]

3. Orientation in closed form. Write $Q = \mathrm{rot}(\theta)\,\mathrm{diag}(1, \eta)$, $\eta = \pm 1$, and (a_i, b_i) for the rows of YQ. For fixed η the overlap residual is $b_i - a_{i+1} = A_i \cos\theta + B_i \sin\theta$, with $A_i = \eta Y_{i2} - Y_{i+1,1}$ and $B_i = -\eta Y_{i1} - Y_{i+1,2}$. Profiling out a common translation centres A and B, and the squared residual becomes the Rayleigh quotient $u^\top H_\eta u$, $H_\eta = \tilde{C}^\top \tilde{C}$, $\tilde{C} = [\tilde{A}\ \tilde{B}]$, $u = (\cos\theta, \sin\theta)$. The eigenvector of the smaller eigenvalue of each 2×2 matrix H_η, with both signs, gives four candidates.
4. Sign selection. Keep the candidate whose regenerated signed RP is closest to R^{obs} in Frobenius norm, both normalised to unit norm.
5. Samples. Correct the translation, $a_i \mapsto a_i - \delta^\star/2$ and $b_i \mapsto b_i + \delta^\star/2$ with $\delta^\star = -\,\mathrm{mean}_i(b_i - a_{i+1})$; set $\hat{x}_1 = a_1$, $\hat{x}_i = \frac{1}{2}(b_{i-1} + a_i)$ for $2 \le i \le T-1$ and $\hat{x}_T = b_{T-1}$, and map their range to the normalised domain.

Orientation costs $O(T)$; the decoder is dominated by the eigendecomposition, $O(T^3)$ in a dense implementation.

4.3. Decoders that isolate the sign and the decoder

Geometric decoder on the unsigned RP. The distances are read directly from the image. Steps 2, 3 and 5 are applied unchanged; step 4 is not available, and the candidate with the smallest overlap residual is kept. On exact distances x and $-x$ fit the overlap equally well, so which one is returned is arbitrary.

Hirata's reconstruction. As the reference for binary RPs we implement Hirata et al. [12, 23]. The unsigned image is thresholded into a binary RP B; an edge joins $i \neq j$ when $B_{ij} = 1$, with length $1 - |G_i \cap G_j|/|G_i \cup G_j|$ and $G_i = \{k : B_{ik} = 1\}$; all-pairs shortest paths give distances and classical MDS

gives the states. Turning the states into T samples is our post-processing: the leading MDS coordinate, the mean appended to reach T samples, and the range map. The method does not prescribe the threshold, and its authors report that it still works with up to 70% of the map set to one [23]; we fixed a recurrence rate of 0.75 in advance and checked it against rates from 0.02 to 0.75 (Section 6.2). On the signed RP the graph is built from $|R|$, so the sign cannot change the recovered geometry; it is used, as in step 4, to choose between the result and its reflection.

Sipers' reconstruction. For unthresholded RPs we also implement the method of Sipers et al. [14]: the two-dimensional Fourier coefficients of the squared RP factor into products of the signal's Fourier coefficients, recovered up to a sign by an eigendecomposition. It assumes a periodic signal and a circular embedding, so it decodes the circular variant of the RP, with the wrap state (x_T, x_1) added and the same 8-bit maps. Frequencies whose power, estimated from the image, is below 10^{-4} of the largest are discarded; this global threshold minimised its own 8-bit error in a pilot, which favours the method (Supplementary Section S5). On the signed RP, the sign selects between the result and its reflection.

Hirata's method thus receives only the binary information left by thresholding, whereas the geometric decoder uses continuous distances.

4.4. GADF decoder

GADF on $[-\alpha, \alpha]$, $\alpha = 0.7$, is decoded in three steps.

1. *Boundary anchor.* The row means are $\mu_i = \frac{1}{T}\sum_j G^{\mathrm{D}}_{ij} = \rho \sin(\phi_i - \psi)$, where $\rho \geq 0$ and ψ lies inside the cone spanned by the sample angles.

That cone is narrower than $\pi/2$, so μ is strictly increasing over it, $e = \arg\max_i |\mu_i|$ is a boundary sample, and $\phi_e = \arccos(-\alpha)$ if $\mu_e > 0$ and $\arccos\alpha$ otherwise.

2. *Relative angles.* For a pivot row p, $\hat{\phi}_j^{(p)} = \phi_e - \arcsin G^{\mathrm{D}}_{ep} - \arcsin G^{\mathrm{D}}_{pj}$.
3. *Pivot averaging.* The angles from the five rows with the smallest $|\mu_p|$ are averaged, $\hat{\tilde{x}}_j = \cos\Big(\frac{1}{5}\sum_p \hat{\phi}_j^{(p)}\Big)$, which reduces the effect of storage.

The decoder is exact in real arithmetic under the hypotheses of Section 3 and costs $O(T^2)$.

4.5. GASF, MTF and XIRP

GASF is decoded with the diagonal inverse of Wang and Oates [1]. For MTF, the T rows are clustered into $Q = 4$ groups, since rows of the same state are equal in a clean MTF; the transition matrix between groups is estimated from the image blocks, the groups are ordered by the Fiedler vector of the Laplacian of its symmetrisation, following spectral seriation [16], and each sample in ordered group k is assigned the bin midpoint $(k - \frac{1}{2})/Q$. The global direction of that order is not determined by the image, and the error of this decoder combines what MTF discards with its own choices; the two are not separated here. XIRP is decoded with the inversion by mean of Hellermann and Lessmann [4].

5. Experimental protocol

5.1. Datasets and windows

Table 2 lists the seven public datasets: human activity (WISDM [17], PAMAP2 [25]), environmental audio (ESC-50 [26]), electrocardiography (PTB-XL [27]), geomagnetism (Swarm [28]), seismology (STEAD [29]) and surface

Table 2: Datasets. Windows of 128 samples available after segmentation; the group is the unit that never appears on both sides of a split.

Dataset	f_s (Hz)	Channels	Task	Windows	Group
WISDM	20	3	5 activities	8 172	subject (51)
PAMAP2	100	9	12 activities	15 085	subject (9)
ESC-50	8 000	1	50 sound classes	624 000	source recording
PTB-XL	100	12	5 classes, multilabel	21 000	patient
Swarm MAG	1	3	none	62 757	–
STEAD	100	3	earthquake vs noise	92 000	event / station
EMG gestures	100	8	6 gestures	880	subject (36)

electromyography (EMG [30]). Every channel is encoded, stored and decoded as its own single-channel image with its own calibration. Recordings are cut into non-overlapping windows of $T \in \{64, 128, 256\}$ samples that never cross a label change or a time gap. Blocks with fewer than 200 windows (EMG at $T = 256$) are not reported.

5.2. Fidelity, robustness and cost

Fidelity uses 500 windows per dataset and length drawn with a fixed seed, and 3 000 for the single-channel ESC-50, so that every block holds between about 1 500 and 6 000 channel series. Writing $r = \max_i x_i - \min_i x_i$, it is measured per channel series by

$$\mathrm{NMAE} = \frac{1}{rT} \sum_i |x_i - \hat{x}_i|,$$

the Pearson correlation, the L_1 distance between unit-mass Welch spectra [18] and the relative energy error; constant series are excluded. Each decoder is run on the floating-point matrix and on the 8-bit image. Robustness adds

zero-mean Gaussian noise of $\sigma \in \{2, 4, 8, 16\}$ intensity levels to every pixel of the stored image, which is then clipped and rounded, with two replicates per level on 200 windows per dataset and length. Cost is the single-thread time to encode, store and decode and the peak traced memory, for T from 64 to 1 024. The sampling-rate study reads 3 000 PTB-XL records at 500 Hz and resamples them with an anti-aliased polyphase filter to 250, 100 and 50 Hz, with windows of 128 samples and, as a control, of a fixed duration of 1.28 s.

5.3. Classification utility

Classifiers are trained on original windows and evaluated on the original and on every reconstruction of the same held-out windows. Up to 6 000 windows per dataset and length are split into three grouped folds that keep every subject, patient, event, station or source recording on one side, stratified by class for single-label data; per-channel standardisation uses training statistics only, and reconstructions are produced independently per window, so no information crosses the split. KNN, an RBF SVM, a random forest, an LSTM, a GRU and a one-dimensional CNN are used with standard settings (Supplementary Section S1). Accuracy, or macro-F1 for the multilabel PTB-XL, is averaged over the three folds. Swarm has no labels. ESC-50 windows of 128 samples span 16 ms of sound, which leaves every classifier near chance, so audio is evaluated on the official ESC-10 subset at 2, 4 and 8 kHz with windows of 512 and 1 024 samples, silent windows removed and accuracy scored per clip.

5.4. Statistical protocol

All comparisons are paired over the same series; differences between decoders are summarised by medians of per-series differences, averaged across datasets with equal weight. A reconstruction is declared *equivalent* to the original for a classifier when the 95% bootstrap interval of the accuracy difference, resampling whole groups [19, 20], lies within ± 0.01, which corresponds to two one-sided tests at the 2.5% level [22, 21]; a non-significant difference is not taken as evidence of equivalence. Equivalence is assessed on single-label data; for PTB-XL only the drop in macro-F1 is reported.

6. Results

6.1. Fidelity and storage (RQ1)

Table 3 gives the 8-bit error of every combination at $T = 128$; the other lengths behave alike (Supplementary Tables S1–S2). Over the 20 reported dataset–length blocks, 61 079 channel series were reconstructed by every combination without decoder failure.

On the floating-point matrix, the geometric decoder on the signed RP, GASF, GADF and XIRP recover the series to machine precision: the median NMAE per block never exceeds 1.4e-13 for the signed RP and 5.6e-17, 8.4e-17 and 2.8e-16 for GASF, GADF and XIRP. Over every series of every block, the largest error is 1.6e-11 for the signed RP and 8.0e-13, 3.6e-16 and 6.3e-16 for GASF, GADF and XIRP. For these four, the 8-bit error is therefore storage error. It stays between 0.0009 and 0.0022 for the signed RP on six datasets and rises to 0.012–0.017 on Swarm; GASF and GADF are of the same order, and XIRP is about an order of magnitude lower. Sipers' reconstruction on the

Table 3: Mean 8-bit NMAE of every encoding–decoder combination at $T = 128$ (500 windows per dataset, 3 000 for ESC-50; every channel). Uns./Sig.: unsigned and signed RP.

	Hirata		Sipers		Geometric (ours)					
Dataset	Uns.	Sig.	Uns.	Sig.	Uns.	Sig.	GASF	GADF	MTF	XIRP
WISDM	0.2411	0.1044	0.1954	0.0043	0.1946	0.0010	0.0015	0.0013	0.2838	0.0002
PAMAP2	0.2345	0.1059	0.1884	0.0050	0.1899	0.0015	0.0016	0.0013	0.2771	0.0002
ESC-50	0.2184	0.0840	0.1916	0.0059	0.1922	0.0017	0.0016	0.0011	0.2740	0.0002
PTB-XL	0.3313	0.1982	0.2680	0.0055	0.2874	0.0016	0.0029	0.0023	0.3404	0.0002
Swarm	0.2778	0.0283	0.2837	0.0045	0.2547	0.0124	0.0026	0.0015	0.2979	0.0002
STEAD	0.2044	0.0868	0.1802	0.0051	0.1771	0.0013	0.0015	0.0011	0.2668	0.0002
EMG	0.1979	0.1205	0.1669	0.0039	0.1680	0.0009	0.0013	0.0022	0.2834	0.0006

circular signed RP comes close, between 0.0028 and 0.0155. Without a node threshold it is exact in floating point but less accurate after storage; with the threshold used here its floating-point error reaches a median of 1.7e-03 per block (Supplementary Section S5).

The remaining combinations reach an NMAE between about 0.03 and 0.34, for two different reasons (Figure 3; spectral and correlational metrics in Supplementary Table S7). On the unsigned RP, the geometric and Sipers' decoders keep the spectrum as well as on the signed RP, because a reflection leaves the spectrum unchanged; its pointwise error and its correlation of about zero come from returning the reflection in about half of the series. Hirata's reconstruction and the MTF representative also distort the spectrum and the energy: they lose the waveform, not only its sign.

6.2. Recurrence plots: encoding versus decoder (RQ2)

Table 3 separates five effects. The spectrum is unchanged by a reflection of the series, so it shows what each decoder keeps of the waveform independently



Figure 3: The windows of Figure 2, decoded from their 8-bit images. Left: the geometric decoder on the signed and on the unsigned RP. Right: Hirata's reconstruction on the unsigned RP and MTF. GASF, GADF and XIRP overlap the original at this scale and are omitted. On Swarm, whose states are nearly collinear, the signed RP deviates visibly at the start of the window.

of its orientation. *Sign.* On the unsigned RP, the geometric decoder keeps the spectrum as well as on the signed RP (PSD-shape distance 0.0007–0.0151, against 0.0034–0.0187), while its correlation with the original stays within ± 0.05: it recovers the shape and loses only the orientation, which the sign restores from the image alone (Figure 3). *Thresholding.* Hirata's reconstruction from the binarised RP distorts the spectrum itself (0.06–0.61): once the image is binary, the waveform is lost, not only its sign. *Continuous distances.* On the same signed RP, the geometric decoder, which uses the distances that thresholding discards, is one to two orders of magnitude more accurate than Hirata's on six datasets; on Swarm the gap narrows to a factor of about two. *Decoder on continuous distances.* Sipers' reconstruction also uses the continuous distances; with the sign it is a median of 0.0015 per series above the geometric decoder, which on the same circular image is 1.8 to 4.6 times more accurate on every dataset (Supplementary Table S5). It is ahead in the benchmark only on Swarm, and the gain comes from the encoding: adding the wrap state raises the median σ_2/σ_1 from 0.0003 to 0.15 and lowers the geometric decoder's error from 0.0122 to 0.0025, below Sipers' 0.0046 on the same image, while on other datasets it changes by at most 0.0002. *Threshold choice.* These comparisons do not depend on the recurrence rate fixed for Hirata: choosing the best rate per dataset with knowledge of the result lowers its error by at most 0.025 on the unsigned and 0.010 on the signed RP, leaving it between 0.02 and 0.31 (Supplementary Table S3).

6.3. Signal characteristics and stability (RQ3)

Remark 1 predicts that storage error is amplified when the states are nearly collinear, and the data agree. None of the 54 040 series with $\sigma_2/\sigma_1 \geq 0.1$

exceeds an 8-bit NMAE of 0.01, whereas 85% of the series with $\sigma_2/\sigma_1 < 0.01$ do. The ratio is an empirical indicator, not a complete explanation: it separates the regimes at the two ends but does not determine which series in between fail. Nor does the share of sign-informative pairs: slow series are informative in nearly every pair, yet their states are the most collinear. Swarm, sampled at 1 Hz, is the limiting case and the only dataset where the signed RP is not in the low-error group.

The sampling rate changes how far apart consecutive samples are, and with it the conditioning (Figure 4b–c). On PTB-XL with windows of 128 samples, from 50 to 500 Hz the median σ_2/σ_1 falls from 0.61 to 0.08, the share of series below 0.1 grows from 0% to 67%, and the 8-bit NMAE of the signed RP rises from 0.0009 to 0.0076, while GASF and GADF stay between 0.0018 and 0.0032 and XIRP is unaffected. A fixed number of samples spans a shorter time at a higher rate, but the trend persists at a fixed duration of 1.28 s, from 0.0012 to 0.0079 (Supplementary Table S4). Sipers' reconstruction rises with the rate as well (Figure 4b), so the effect belongs to the RP, not to one decoder. The resampled and the distributed 100 Hz files give the same error (0.0015 and 0.0015). Resampling also changes the spectral content of each window, and the study covers one dataset, so the trend is attributable to the rate as realised by anti-aliased resampling of PTB-XL, not to the rate in isolation.

6.4. Classification utility (RQ4)

Six classifiers, five labelled datasets and three lengths give 84 combinations on 72 917 windows (Table 4). On the original windows, accuracy ranges from 0.17 to 0.81 and PTB-XL macro-F1 from 0.14 to 0.61; the lowest values

(KNN on EMG) are near chance, where drops are less informative. The four combinations that are exact in floating point never lose more than 0.007 and are equivalent within ± 0.01 in nearly every single-label combination; where equivalence is not established, the observed difference is still below 0.01 and the interval is too wide rather than the drop large. Sipers' reconstruction with the sign joins this group, with a largest drop of 0.008, but is equivalent in fewer combinations (59 of 66, against 63 for the geometric decoder), consistent with its larger 8-bit error. Without the sign, Sipers' reconstruction, Hirata's on either encoding, the geometric decoder on the unsigned RP and MTF lose between 0.08 and 0.12 on average and up to 0.40: whichever decoder is used, the unsigned RP loses the orientation and with it the accuracy. The same separation holds per clip on ESC-10 (18 000 windows, 36 combinations), where the exactly decodable combinations lose at most 0.017.

6.5. Robustness (RQ3) and cost (RQ4)

With noise of $\sigma = 16$ intensity levels on the stored image, the mean NMAE at $T = 128$ rises to 0.011 for XIRP, 0.041 for the signed RP, and 0.064 and 0.064 for GASF and GADF, and 0.083 for Sipers' reconstruction on the signed RP (Figure 4a). The geometric decoder on the signed RP degrades less than both angular fields on 6 of the 7 datasets, with the three level on Swarm, and less than Sipers' on 7 of the 7. The lossy combinations move by at most 0.020, because their error is already dominated by what is discarded before storage.

Hirata's reconstruction has the steepest cost, with an exponent of about 2.6 due to all-pairs shortest paths (Supplementary Table S6). The geometric decoder on the signed RP is 2.9 times faster at $T = 256$ and 5.2 times faster at $T = 1024$, and costs about as much as Sipers' reconstruction at $T = 256$

Table 4: Drop from the original accuracy (macro-F1 on PTB-XL) for classifiers trained on original windows and tested on reconstructions. $|\Delta| \leq 0.01$: share of combinations whose observed drop is within 0.01. Equivalent: 95% group-bootstrap interval within ± 0.01, over the single-label combinations.

	Five datasets, $T \in \{64, 128, 256\}$				ESC-10, per clip	
Decoder	Mean drop	Max drop	$\lvert\Delta\rvert \leq 0.01$	Equivalent	Mean drop	Equivalent
Hirata / uns.	0.120	0.376	19%	11/66	0.071	0/36
Hirata / sig.	0.080	0.283	26%	12/66	0.060	0/36
Sipers / uns.	0.098	0.371	19%	13/66	0.021	0/36
Sipers / sig.	0.000	0.008	100%	59/66	0.001	28/36
Ours / uns.	0.099	0.359	21%	13/66	0.024	0/36
Ours / sig.	0.000	0.007	100%	63/66	0.001	33/36
GASF	-0.000	0.005	100%	63/66	0.001	34/36
GADF	-0.000	0.002	100%	63/66	0.000	32/36
MTF	0.099	0.400	19%	7/66	0.106	0/36
XIRP	0.000	0.002	100%	65/66	-0.000	36/36

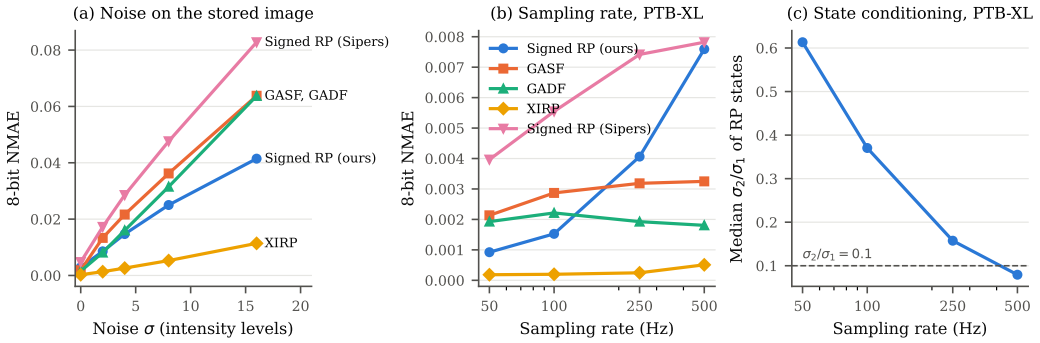


Figure 4: (a) Mean 8-bit NMAE against Gaussian noise added to the stored image ($T = 128$, mean over the seven datasets, two replicates per level). (b) 8-bit NMAE and (c) median conditioning σ_2/σ_1 of the RP states against the sampling rate (PTB-XL, $T = 128$, anti-aliased resampling from 500 Hz). Hirata's reconstruction, the geometric and Sipers' decoders on the unsigned RP and MTF stay between 0.08 and 0.34 throughout and are omitted.

and half as much at $T = 1024$. GASF, GADF, XIRP and MTF are one to two orders of magnitude cheaper. Memory grows quadratically for all of them.

7. Discussion

What each representation retains. Three groups emerge, consistent across domains, lengths and classifiers. The signed RP with the geometric decoder, GASF, GADF on its restricted range and XIRP are exact in floating point under the stated conditions, so their error after storage is storage error, too small to affect classifiers trained on original signals. The unsigned RP determines the shape of the series but not its sign. The binary RP and MTF are many-to-one, so no decoder can be exact for every series, and the evaluated decoders lose most of the waveform; better approximations, for instance with learned priors, are not ruled out, but they cannot be read back

with a guarantee of matching the encoded series.

What the sign and the decoder add. RP reconstruction needs both. The sign fixes the orientation that the unsigned RP leaves open, and one sign-informative pair suffices (Proposition 1). The continuous distances that thresholding discards account for the advantage over Hirata's reconstruction, and among decoders that use them the geometric one is more accurate and more robust than Sipers' on the same image, without assuming a periodic signal or choosing a frequency threshold.

When recovery becomes unstable. The signed RP is the only exactly decodable representation whose stability depends strongly on the signal: slowly varying windows, from Swarm at 1 Hz or from an ECG at 500 Hz, have nearly collinear states that amplify storage error, for Sipers' decoder as for ours.

Using the benchmark. The comparison is not a ranking, since a model may need a particular image. With recurrence plots, the signed RP and the geometric decoder give a route back to the series, with sufficient conditions for exactness, a measurable indicator of stability and less sensitivity to pixel noise than the angular fields on most datasets. GADF can be decoded if its angular range is restricted at encoding time; MTF and the binary RP give only approximations. Where the image is free to choose and the signal is sampled far above its bandwidth, the angular fields and XIRP are cheaper and independent of its conditioning. XIRP is an explicitly invertible reference: the redundancy of its column proposals plausibly explains its lower error, which says nothing about whether its images are better inputs for classification or generation.

Limitations. Images are deterministic outputs of known encoders; pixel noise measures local stability only. Physical units need the per-channel calibration, which a generative setting would have to supply; only 8-bit storage and independent channels are studied. PTB-XL and ESC-50 are split by patient and recording, not by official folds. Hirata's reconstruction is our implementation, with a threshold the method does not prescribe, and Sipers' runs on the circular RP it requires, with a node threshold chosen in a pilot. The sampling-rate study covers one dataset, and σ_2/σ_1 is an empirical indicator, not an error bound.

8. Conclusions

We measured, under one framework, how much of a time series survives its image for the most widely used image representations and XIRP, on seven datasets from six domains. Four are exactly decodable in floating point under stated conditions and, after 8-bit storage, keep what classifiers trained on original signals need: the signed recurrence plot with a geometric decoder, GASF on $[0, 1]$, GADF on a restricted angular range, and XIRP. The unsigned recurrence plot is recovered only up to sign; the binary recurrence plot and the Markov transition field, which are many-to-one, lose most of the waveform with the decoders evaluated. For recurrence plots, exact recovery needs the sign and a decoder that uses the continuous distances; stability after storage depends on how collinear the states are, and hence on the sampling rate. Images produced by generative models are the next step.

CRediT authorship contribution statement

Adriano García-Giralda Milena: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Validation, Visualization, Writing – original draft, Writing – review & editing. **Francisco García Moreno:** Conceptualization, Methodology, Supervision, Writing – review & editing. **José Antonio Jiménez Madrid:** Methodology, Formal analysis, Writing – review & editing. **María Bermúdez Edo:** Supervision, Writing – review & editing.

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Data availability

All datasets are public and cited. Code, manifests, configurations and derived results: `https://github.com/adrianoggm/TFG-Tratamiento-de-Series-Temporales` [pending: DOI, #137].

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

[pending: tools and uses, #138.] All AI-assisted content was critically reviewed, verified and revised by the authors, who take full responsibility for the accuracy, originality and integrity of the manuscript.

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